

QBUS2820 Assignment 1 (25 marks)

August 28, 2026

1 Background

Access to suitable and affordable rental housing is an important issue for households, property managers, and urban planners. Accurate prediction of weekly rent can help property managers assess rental listings and can support analysis of how housing characteristics and location are associated with rental costs.

In this assignment, you take the role of a data analyst working for a rental property platform in Harbour City, a fictional large Australian metropolitan area. Your task is to develop a regression model that predicts the advertised weekly rent of a residential property. The response variable, **WeeklyRent**, in the dataset **WeeklyRent_training.csv** records the advertised rent of a property in Australian dollars per week. A description of the response and covariates is provided in Table 1.

Variable	Description
WeeklyRent	Advertised residential rent in Australian dollars per week.
DistanceCBD	Distance from the property to the Harbour City central business district, in kilometres.
Bedrooms	Number of bedrooms in the property.
FloorArea	Internal floor area of the property, in square metres.
PropertyAge	Age of the property, in years.
NearTrain	Equals 1 if the property is within approximately 1 km of a train or metro station, and 0 otherwise.
Furnished	Equals 1 if the property is advertised as furnished, and 0 otherwise.
HighDemandArea	Equals 1 if the suburb was classified as a high-demand rental area using historical vacancy and local-amenity indicators, and 0 otherwise.

Table 1: Description of variables.

Your task is to build a regression model that predicts **WeeklyRent** using these covariates. You are also provided with **WeeklyRent_test_noLabel.csv**, which contains the test covariates but does not contain **WeeklyRent**. The complete test dataset, **WeeklyRent_test.csv**, has the same structure as the training data and will be available only to the marker.

1.1 Test Error

Prediction accuracy will be measured using mean squared error (MSE) on the test data. Let $\hat{y}_i$ be the prediction of y_i , where y_i is the i -th value of **WeeklyRent** in the test data. The test error is

$$\text{Test error} = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} (\hat{y}_i - y_i)^2,$$

where n_{test} is the number of observations in the test dataset.

2 Submission Instructions

Please submit the following **THREE** files through Canvas. You may submit additional files if they are required by your implementation.

1. A report named **SID_Assignment1_document.pdf**, where “SID” is replaced by your student ID.
2. A Jupyter Notebook named **SID_Assignment1_implementation.ipynb** that implements your data analysis and produces the test error when run by the marker.
3. A CSV file named **SID_Assignment1_prediction.csv** containing predictions for `WeeklyRent_test_noLabel.csv`. The file must contain **exactly one column** named `WeeklyRent`, with predictions in the same row order as the test data.

If additional files are required, use your student ID and a short descriptive filename.

2.1 Report Requirements

Your report must:

- explain how exploratory data analysis was conducted;
- describe the prediction methods and predictors considered;
- explain and justify your modelling decisions;
- report and discuss relevant numerical results to four decimal places;
- present relevant graphs and tables clearly; and
- contain enough detail for another data scientist to understand and reproduce your analysis.

The report must be within **15 pages**, including appendices, figures, tables, and the AI-use statement. Use a standard font size of 11 or 12 points.

The report must be written separately using Word, LaTeX, or similar document software. A Jupyter Notebook exported directly to HTML or PDF will not be accepted as the report. The report should be clearly structured, easy to read, and focused on communicating your analysis rather than collecting raw output.

2.2 Notebook Requirements

Assume that `WeeklyRent_training.csv` and `WeeklyRent_test_noLabel.csv` are in the same folder as your notebook. The marker will place the complete file `WeeklyRent_test.csv` in that folder before running the notebook.

The following code must appear in the **last code cell** of your submitted notebook:

```
import pandas as pd

WeeklyRent_test = pd.read_csv("WeeklyRent_test.csv")

# YOUR CODE HERE: code that produces the test MSE test_error

print(test_error)
```

The final cell must calculate the same test MSE that you would obtain if you were given the complete test data. Since you do not have access to `WeeklyRent_test.csv`, do not execute this final cell before submission. All earlier cells must run without errors and must create any fitted model, predictors, or objects required by the final cell.

Before submission, restart the kernel and run all student-executable cells from the beginning, leaving only the marker-only final cell unexecuted. Your notebook must include clear explanations and comments so that the marker can understand and run your approach.

Use only methods covered in the lectures and tutorials. You may use publicly available Python libraries to implement those methods.

2.3 Acknowledging the Use of AI

You must include a statement at the end of your report indicating whether or not generative AI tools were used.

- If you did not use generative AI tools, state clearly that no AI assistance was used.
- If you used generative AI tools, state the name and version of each tool, its publisher, and how it was used in completing the assignment.

Up to 2 marks may be deducted if the required statement is missing.

You are also required to maintain a record of your AI use during the assignment and retain copies of relevant AI-generated outputs as evidence of your research and writing process.

3 Marking Criteria

This assignment is worth 25 marks. Fourteen marks are allocated to the report and 11 marks to the Python implementation.

1. Prediction accuracy and implementation (11 marks).

The marker will run `SID_Assignment1_implementation.ipynb`. The test MSE will be compared with the smallest test MSE among all submissions, including models developed by the teaching team.

- If the notebook runs successfully and produces a valid test MSE, up to 11 marks will be awarded based on prediction accuracy and the appropriateness of the implementation.
- If the notebook cannot be run successfully or does not produce a valid test MSE, partial marks of at most 3 may be awarded based on the appropriateness of the submitted implementation.

You are responsible for ensuring that your notebook is executable. Failure to produce a valid test MSE may result in the loss of up to 8 marks allocated to prediction accuracy.

2. Report (14 marks).

Marks will be awarded for the appropriateness of the prediction method and for the quality, detail, discussion, and explanation of the data analysis. Refer to the detailed marking criteria provided on Canvas.

3. Prediction CSV.

Up to 2 marks may be deducted if the prediction CSV is missing or does not follow the required filename, column name, length, or row order.

4 Special Consideration

If you have been granted special consideration with an extended due date later than the original closing date, your new closing date will be the same as your extended due date. No submission will be accepted after that date. Failure to submit by the extended due date will result in a mark of zero.

5 Final Notes

Read all instructions carefully and seek clarification early if you are unsure about any requirement. If you believe there is an error in the assignment, contact the teaching team as soon as possible.